FACILITATION GUIDE

LESSON TITLE: Operating Systems Written Exam

LESSON DESIGNATOR: DS.24.02

TOTAL LESSON TIME: 2 Hours

LEARNING OBJECTIVES:

TLOs:

1. Given a command's mission, threat assessment, communications plan, required and configuration equipment, installation media, cybersecurity resource references, and personnel. Implement Data Systems to establish data systems and services in accordance with applicable STIGs and local cybersecurity policies in support of the communications plan. (0671-IMPL-1002).
2. Given a command's mission, threat assessment, communications plan, required and configured equipment, install media, cybersecurity resource references, and personnel. Operate Data Systems to maintain secure data systems and services that can be operated in accordance with applicable STIGs and local cybersecurity policies in support of the communications plan. (0671-OPER-1002).

ELOs:

1. Without the aid of reference, identify in writing the characteristics of operating systems, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002a).
2. Without the aid of reference, identify in writing the client operating system hardware requirements, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002ab).
3. Without the aid of reference, identify in writing the characteristics of replication, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002aj).
4. Without the aid of reference, identify in writing the characteristics of forests, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002ar).
5. Without the aid of reference, identify in writing the characteristics of active directory domain services, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002as).
6. Without the aid of reference, identify in writing the characteristics of domain controllers, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002av).
7. Without the aid of reference, identify in writing the characteristics of group policies, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002ax).
8. Without the aid of reference, identify in writing the characteristics of active directory sites, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002ay).
9. Without the aid of reference, identify in writing the characteristics of server backup utilities, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002ba).
10. Without the aid of reference, identify in writing the characteristics of workstation clients, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bc).

11. Without the aid of reference, identify in writing the client operating system deployment options, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bd).
12. Without the aid of reference, identify in writing the active directory backup types, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bn).
13. Without the aid of reference, identify in writing the characteristics of performance monitoring, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bp).
14. Without the aid of reference, identify in writing the characteristics of domains, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bq).
15. Without the aid of reference, identify in writing the characteristics of domain trees, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002br).
16. Without the aid of reference, identify in writing the characteristics of domain name services, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bu).
17. Without the aid of reference, identify in writing the domain name services record types, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002bv).
18. Without the aid of reference, identify in writing the characteristics of administrative tools, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002cw).
19. Without the aid of reference, identify in writing the characteristics of domain services, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002e).
20. Without the aid of reference, identify in writing the electronic messaging solutions, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002eo).
21. Without the aid of reference, identify in writing the characteristics of messaging servers, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002ep).
22. Without the aid of reference, identify in writing the unified messaging concepts, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002eq).
23. Without the aid of reference, identify in writing the characteristics of messaging service recipients, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002er).
24. Without the aid of reference, identify in writing the types of messaging systems, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002es).
25. Without the aid of reference, identify in writing the messaging server hardware requirements, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002et).
26. Without the aid of reference, identify in writing the messaging server software requirements, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002eu).
27. Without the aid of reference, identify in writing the messaging server roles hardware requirements, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002ev).

28. Without the aid of reference, identify in writing the characteristics of messaging servers on a domain, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002ew).
29. Without the aid of reference, identify in writing the messaging server domain requirements, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002ex).
30. Without the aid of reference, identify in writing the messaging server roles, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002ey).
31. Without the aid of reference, identify in writing the characteristics of a command-line messaging server installation, in accordance WITH ISBN: 978-1-119-23205-6. (0671-IMPL-1002ez).
32. Without the aid of reference, identify in writing the characteristics of mail-enabled groups, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fa).
33. Without the aid of reference, identify in writing the characteristics of an exchange administration center, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fb).
34. Without the aid of reference, identify in writing the client access solutions, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fc).
35. Without the aid of reference, identify in writing the characteristics of messaging clients, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fd).
36. Without the aid of reference, identify in writing the characteristics of messaging systems, in accordance with ISBN: 978-1-119-23205-6. (0671-IMPL-1002fg).
37. Without the aid of reference, identify in writing the definition of operating systems, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002i).
38. Without the aid of reference, identify in writing the purpose of operating systems, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002j).
39. Without the aid of reference, identify in writing the types of operating systems, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002k).
40. Without the aid of reference, identify in writing operating system applications, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002l).
41. Without the aid of reference, identify in writing operating system hardware requirements, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002m).
42. Without the aid of reference, identify in writing the definition of a server, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002n).
43. Without the aid of reference, identify in writing the types of servers, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002o).
44. Without the aid of reference, identify in writing the server operating system hardware requirements, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002p).

45. Without the aid of reference, identify in writing the server operating system installation options, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002q).
46. Without the aid of reference, identify in writing the characteristics of server updates, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002r).
47. Without the aid of reference, identify in writing the updated server hardware requirements, in accordance with ISBN: 978-1-119-35914-2. (0671-IMPL-1002s).

REFERENCES:

Reference Title	Publication ID
Defense Information Systems Agency (DISA) Security Technical Implementation Guides (STIGs) Library	DISA STIGs Library
DISA Security Technical Implementation Guides	DISA STIGS
MASTERING MICROSOFT EXCHANGE SERVER 2016	ISBN: 978-1-119-23205-6
MCSA WINDOWS 2016 COMPLETE STUDY GUIDE	ISBN: 978-1-119-35914-2

PERSONNEL REQUIRED: 3 facilitators; 1 Course Chief, 1 Curriculum Developer

FACILITIES REQUIRED: Bldg 1763 Room 250

MATERIALS/EQUIPMENT:

Presentation Software PS	Instructors Intelligence Workstation (I) IW
Student Intelligence Workstation (S) IW	Admin Time AD
Check-in CHKIN	Checkout CKOUT
Computer Workstation w/ Software & Network Access CWS(N)	Dry Erase Board DB
End of Course Critique EOCC	Examination EXAM
Graduation Certificate Grad Cert	Learning Management System LMS
Lesson Plan LP	Laptop Computer LTC
Performance Evaluation Checklist PECL	Performance Exam PERF EXAM
PowerPoint Presentation PPT	Reference Materials (Digital and/or Paper Based) REF
Smart Board SB	Student Handout SH
Tablet TAB	Training Schedule TS

LEARNING EXPERIENCE ACTIVITIES:

Method	Code
Academic Orientation	ACOR
Administrative	ADMIN
Coaching	CCH

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Course Introduction	CI
Check-in	CKIN
Check-Out	CKOUT
Demonstration	D
End of Course Critique	EOCC
Guided Discussion	GD
Graduation	GRAD
Gallery Walk	GW
Homework	HW
Informal Lecture	L
Informal Lecture/Guided Discussion	IL/GD
Informal Lecture/Gamification	IL/GM
Informal Lecture/Jigsaw	IL/JSW
Informal Lecture/Questioning	IL/Q
Independent Study Time	IST
Leadership Guided Discussion	LDGD
Practical Application	PA
Quiz	Q
Socratic Questioning	SOQ
Teach Back	TB
Think-Pair-Share	TPS
Performance Exam	X(P)
Exam Review	X(RE)
Written Examination	X(W)

Learner Role(s): **Data Systems Administrator**

The Data Systems Administrator is a critical role in the 21st century digital landscape, responsible for designing, implementing, and managing data systems that support organizational operations. This role requires a deep understanding of the fundamentals of information technology, as well as the ability to apply 21st century learning concepts such as critical thinking, problem-solving, and collaboration.

Key Responsibilities:

- Design and implement data systems that meet organizational needs and requirements.
- Manage and maintain data systems, including hardware, software, and networks.
- Troubleshoot and resolve data system issues, using critical thinking and problem-solving skills.
- Collaborate with stakeholders to identify and prioritize data system needs and requirements.
- Develop and implement data backup and recovery plans, ensuring data integrity and security.
- Stay up to date with emerging trends and technologies in data systems administration, applying 21st century learning concepts such as innovation, creativity, and adaptability.

21st Century Learning Concepts:

- Critical thinking: analyzing complex data system issues and developing effective solutions.
- Problem-solving: troubleshooting and resolving data system problems, using creative and innovative approaches

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- Collaboration: working with stakeholders to identify and prioritize data system needs and requirements
- Communication: effectively communicating data system information and requirements to stakeholders
- Creativity: designing and implementing innovative data systems solutions that meet organizational needs
- Innovation: staying up-to-date with emerging trends and technologies in data systems administration, and applying them to improve organizational operations

Learning Objectives:

- Understand the fundamentals of information technology, including data systems, networks, and cybersecurity.
- Analyze complex data system issues and develop effective solutions, using critical thinking and problem-solving skills.
- Design and implement data systems that meet organizational needs and requirements, using collaboration and communication skills.
- Troubleshoot and resolve data system issues, using creative and innovative approaches.
- Develop and implement data backup and recovery plans, ensuring data integrity and security.
- Stay up-to-date with emerging trends and technologies in data systems administration, applying 21st century learning concepts such as innovation, creativity, and adaptability.

Performance Tasks:

- Design and implement a data system for a fictional organization, using critical thinking and problem-solving skills.
- Troubleshoot and resolve a complex data system issue, using creative and innovative approaches.
- Develop and implement a data backup and recovery plan, ensuring data integrity and security.
- Collaborate with stakeholders to identify and prioritize data system needs and requirements, using communication and collaboration skills.
- Create a presentation or report on emerging trends and technologies in data systems administration, applying 21st century learning concepts such as innovation, creativity, and adaptability.

Resources:

- Online learning platforms and resources, such as video tutorials and interactive simulations
- Textbooks and e-books on data systems administration and information technology
- Industry publications and research articles on emerging trends and technologies in data systems administration

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- Collaboration tools, such as discussion boards and group projects, to facilitate communication and teamwork.

Facilitator Role: The Data Systems Administration Instructor plays a crucial role in delivering instruction and guiding learners in developing knowledge and skills related to data systems administration, including hardware, software, networks, and cybersecurity. To achieve this, the instructor employs a range of instructional methods, including lectures, discussions, and hands-on activities, to engage learners and promote understanding. By facilitating group discussions, providing individualized support, and using technology such as virtual labs and online resources, the instructor creates a positive and inclusive learning environment that encourages active learning and participation. The instructor's key responsibilities include delivering instruction, guiding learners in completing performance tasks and projects, and evaluating learner progress, all while ensuring learners understand the fundamentals of information technology and can apply their knowledge to troubleshoot and resolve data system issues. Ultimately, the instructor's effectiveness is measured by learner engagement, understanding, and application of data systems administration concepts, as well as their ability to adapt instruction to meet learner needs and promote understanding.

LEARNING EXPERIENCE REFLECTION: Describe your reflection method. What feedback will you provide?

As I reflect on my learning experience, I will provide feedback on the following aspects:

- The effectiveness of the instructional methods and materials used in the course.
- The clarity and relevance of the learning objectives and outcomes
- The level of support and guidance provided by the instructor and peers.
- The opportunities for practice and application of the concepts and skills learned.
- The overall quality of the learning experience and its impact on my knowledge and skills in Data Systems Administration

I will provide constructive feedback that is specific, timely, and respectful, with the goal of improving the learning experience for future learners. My feedback will include suggestions for improvements, such as additional resources or support, and recommendations for how to build on the strengths of the course. By providing thoughtful and reflective feedback, I aim to contribute to the ongoing improvement and refinement of the Data Systems Administration course, and to help ensure that future learners have a high-quality and effective learning experience.

ASSESSMENT: How do learners demonstrate their understanding?

- Quizzes and exams to assess understanding of the fundamentals of information technology.

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- Performance tasks and projects to assess application of critical thinking, problem-solving, and collaboration skills.
- Peer review and feedback to assess communication and collaboration skills.

SAFETY:

☐ Review/Reassess the learning environment/location for changes that affect the original Risk Assessment.

APPROVING SIGNATURE _____DATE _____